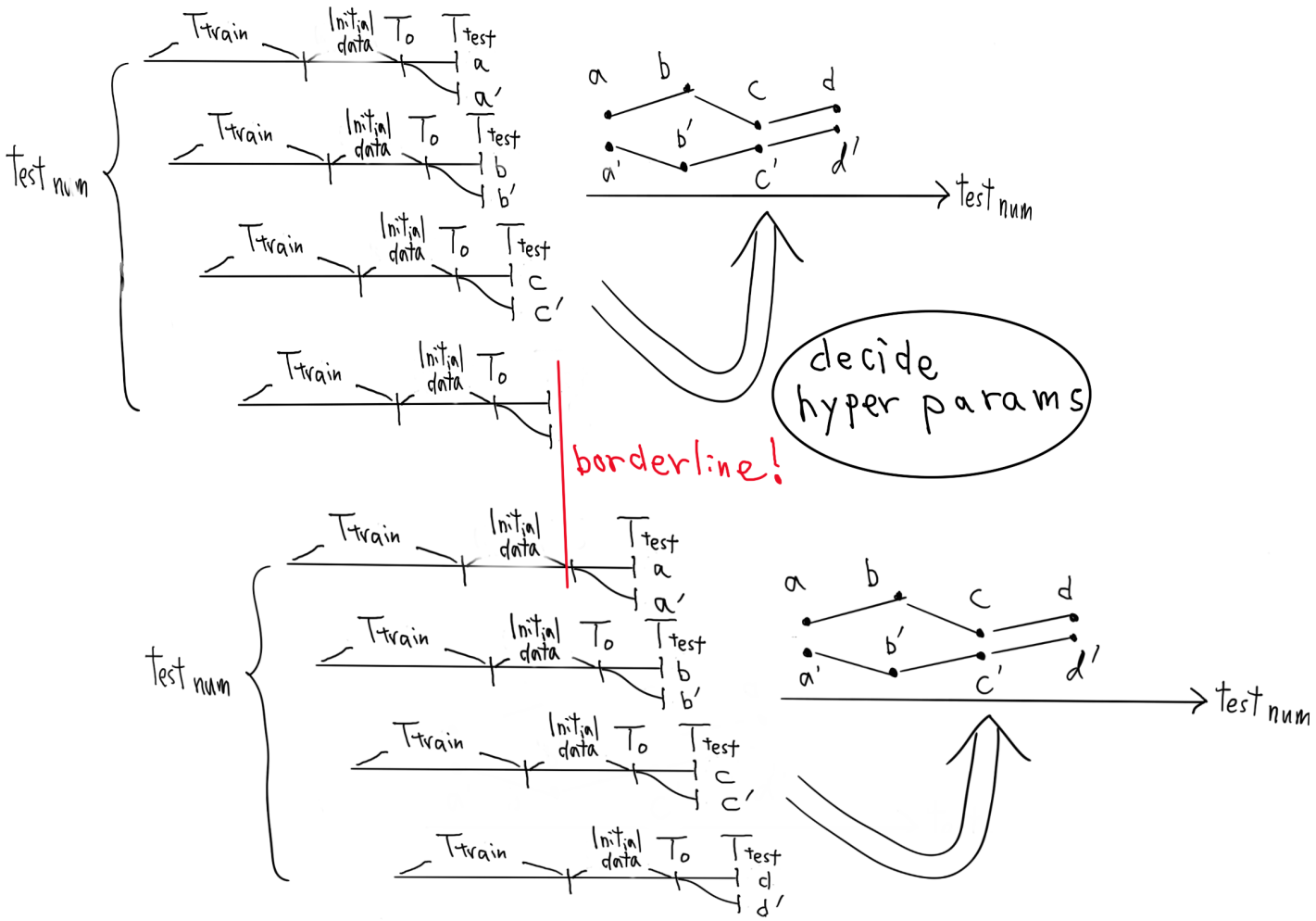
ML Results
(Reservoir computing)

Learning framework: ElNino and MJO

Pearson correlation coefficient (PCC) between real data and prediction



Generalization performance evaluation: $T_{test} = 1, 2, \cdots$

Threshold of the correlation is 0.5 in this competition!!

El Nino prediction (Jinno-Mitsui-Nakai-Saiki-Y 2025)

Nino 3.4 (this time series expresses the strength of El Nino)

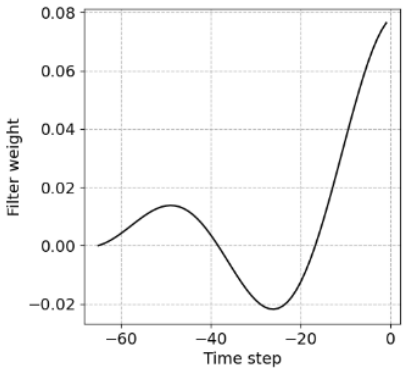
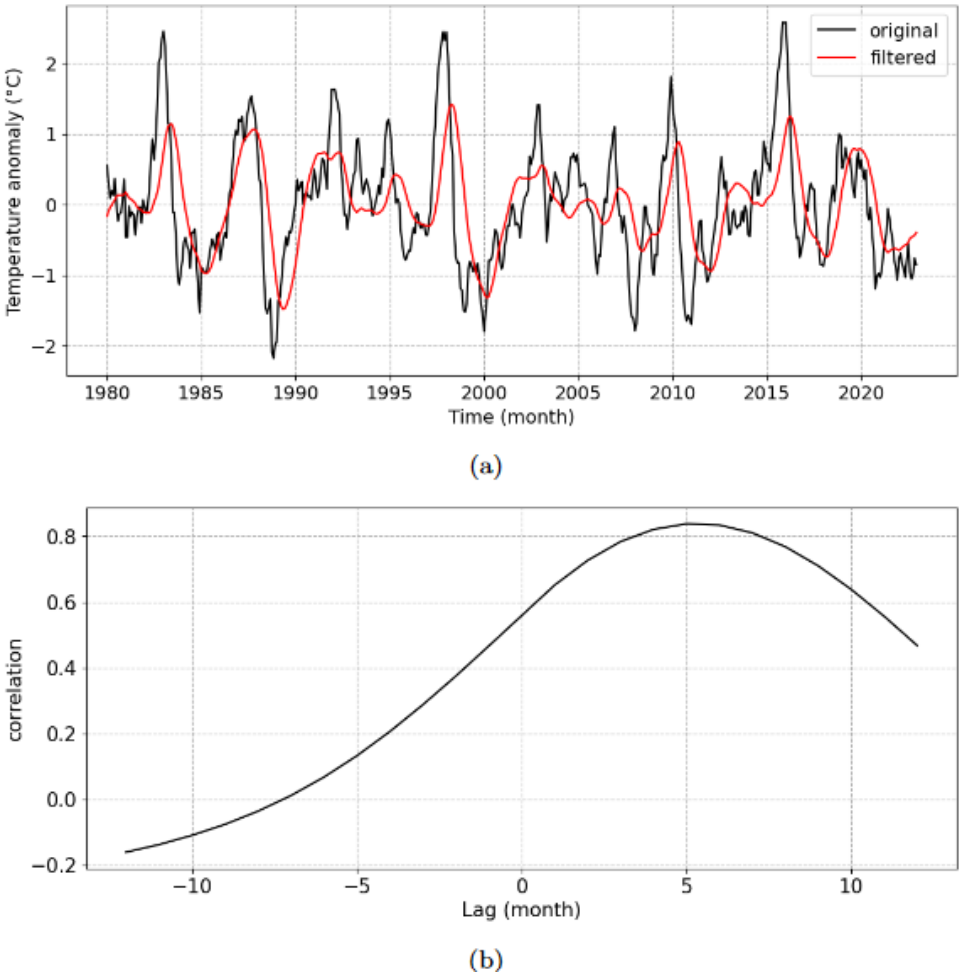


Figure: data-driven filter



(a): original (black) and filtered (red)

(b): Lag correlation between original and filtered : correlation > 0.8

Result (El Nino)

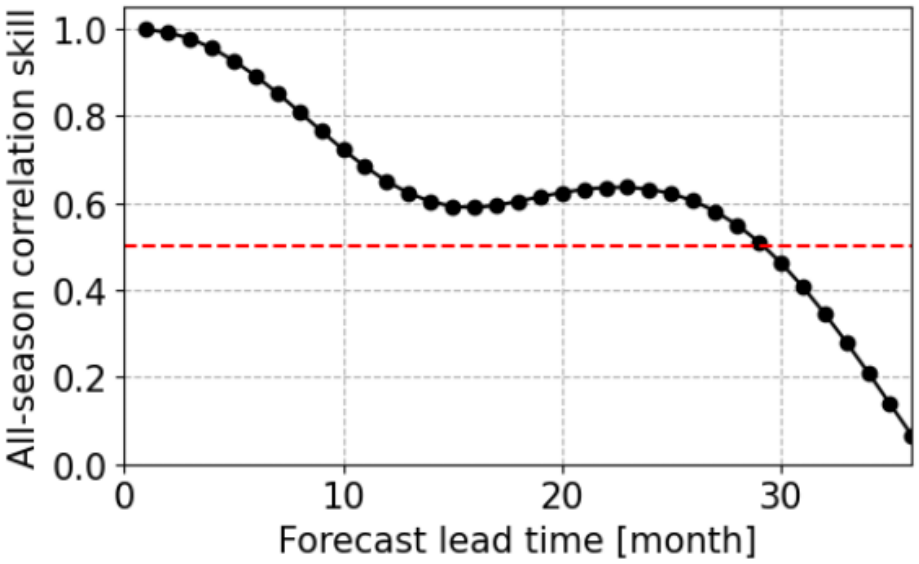


Figure: Prediction with our novel filter
(29 minus 5(lag)$=24$ months)

- Reservoir Hassanibesheli-Kurths-Boers (2022): 19 months (but incorporating future values due to the band-pass filter)
- CNN Ham-Kim-Luo (2019): 17 months
- Physical model SINTEX-F: 12 months

Result (MJO prediction)

Nakai-Suematsu-Miura-Y-Takasuka-Jinno-Saiki (arXiv)

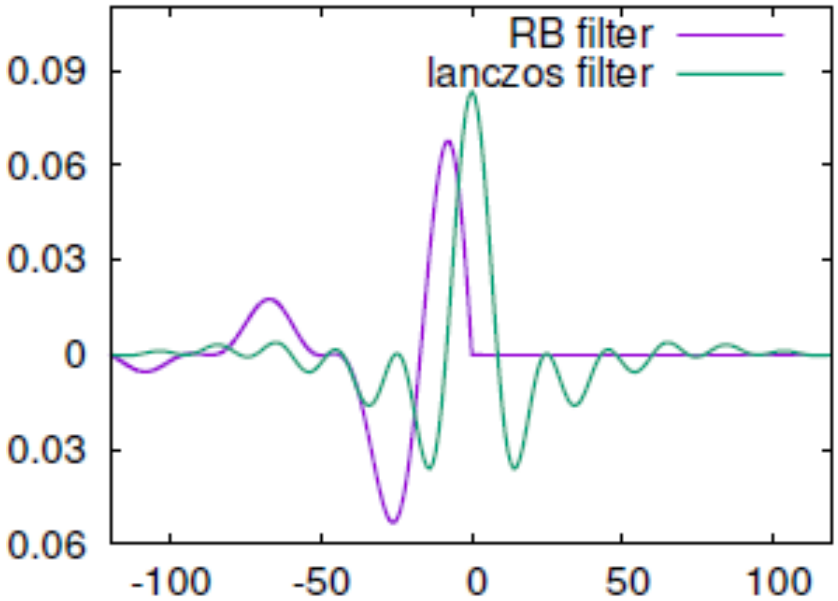


Figure: Lag (weighted center): -7.9

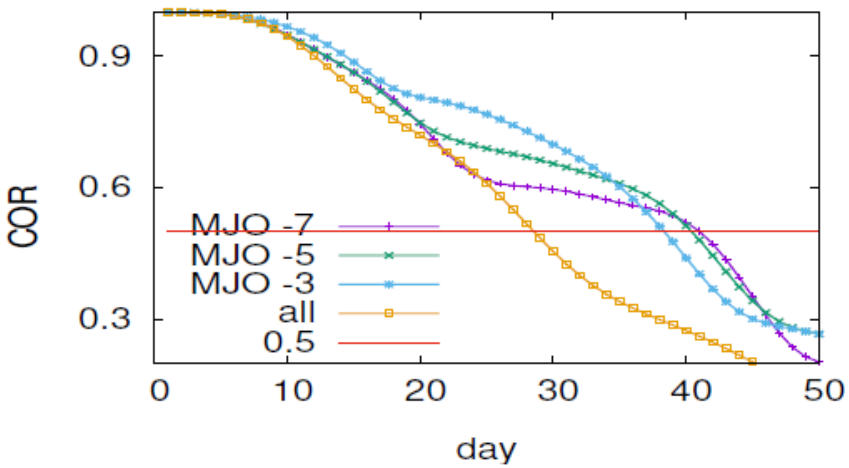
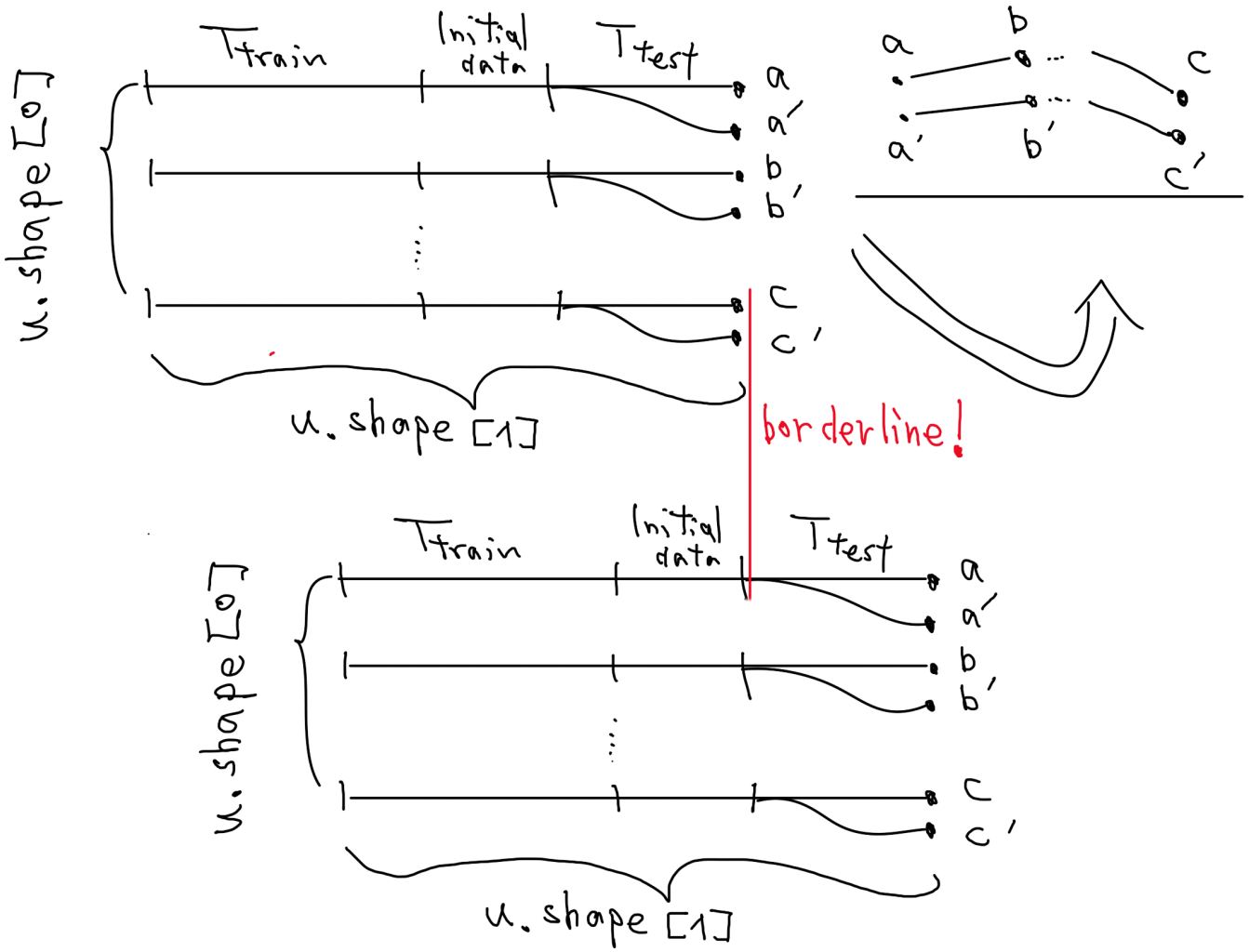


Figure: Pred with our RB filter (29 minus 8(lag)=21)

Even when using state-of-the-art physical models with high computational cost, the prediction accuracy is only on the order of about 20 days!

Learning framework: Tokyo WindSpeed and SP500

Mean absolute error (MAE) between real data and prediction

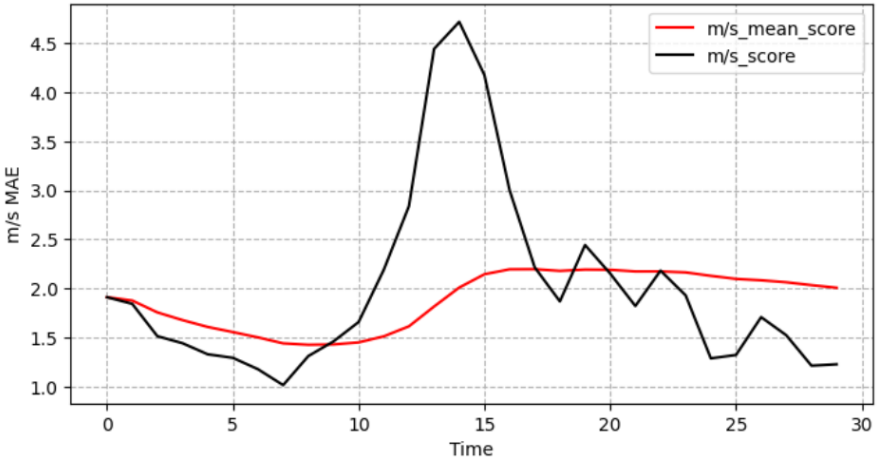


Generalization performance evaluation: choose one T_{test}

Result (windspeed in Tokyo region)

- predict 3 hours ahead ($T_{test} = 3$)
- most recent 1,000 hours of data are used to train($=: T_{train}$)
- Thirty independent trials were performed using a rolling-window validation scheme

Result MAE: 2.01 m/s (data-driven filter is applied: correlation 0.83, Jinno-Mitsui-Nakai-Saiki-Yoneda 2025)



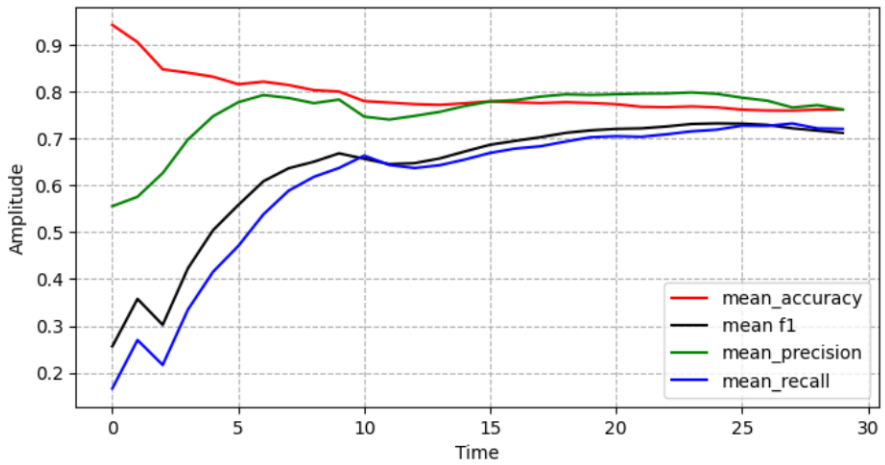
Cheng and Steenburgh (2005) CIRP, WRF (physical models)

MAE: 1.6 $\sim$ 1.9 m/s, but unclear whether a filter is applied...

Result (stock S&P500)

- predict 1 day ahead ($T_{test} = 1$)
- most recent 60 days of data are used to train($=: T_{train}$)
- learnable params $\approx$ 1,000
- Thirty independent trials were performed using a rolling-window validation scheme

Result Accuracy 0.762, F1 0.712, Recall 0.721 (data-driven filter is applied: correlation 0.985, Jinno-Mitsui-Nakai-Saiki-Yoneda 2025)

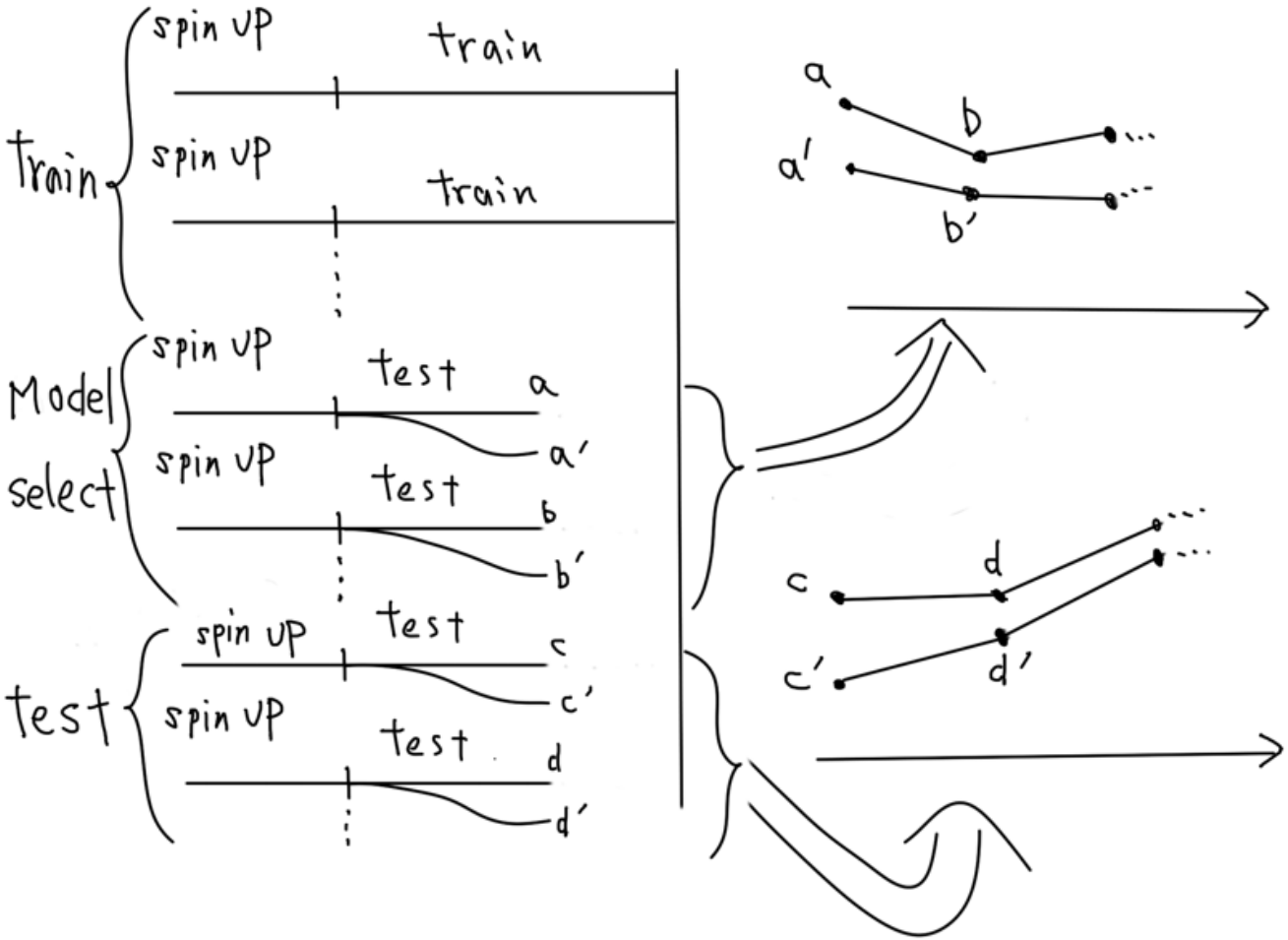


Gil, Duhamel-Sebline, McCarren (2024) xLSTM-TS

Accuracy 0.709, F1 0.730, Recall 0.768, wavelet filter is applied.
learnable params = 125,389, very huge!!

Learning framework: Lorenz 63

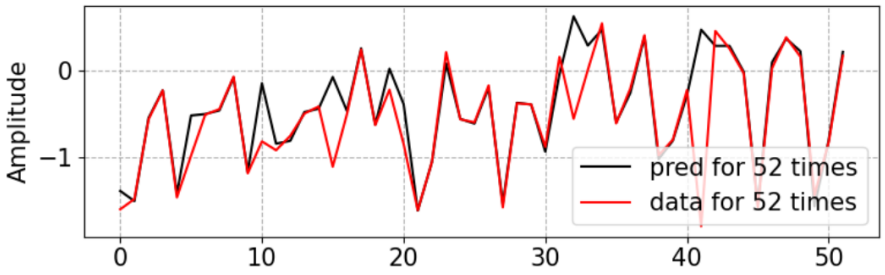
Mean absolute error (MAE) between real data and prediction



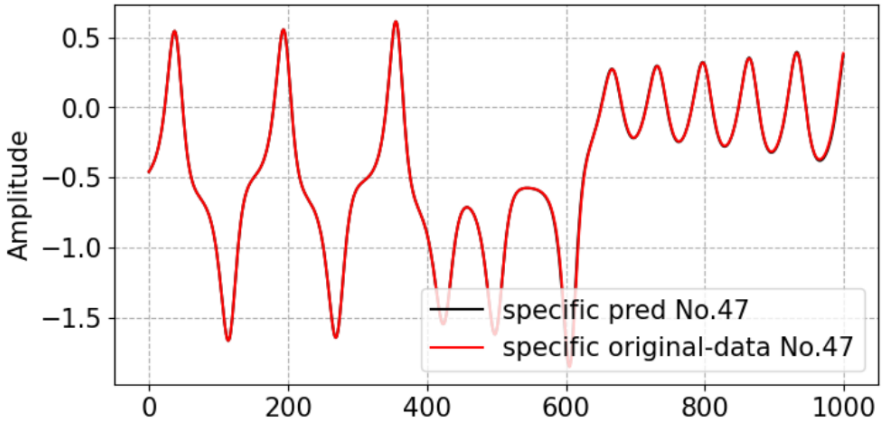
Generalization performance evaluation: take average between 1 and T_{test}

Result (Lorenz system)

- training dataset$\approx 600,000 \times 3$
- predict 1000-steps ahead ($T_{test} = 1000$), nondimensional time of 10
- standardized dataset
- MAE as the mean error over the entire prediction horizon



1000-step-ahead pred vs. ground truth



ParallelReservoir: mean-MAE 0.014,
Prediction: 1000 steps ahead

Feng et al. (2025) Physics-Guided Learning (PGL) method

A standardized mean-MAE of approximately $0.10 \sim 0.12$.

- training dataset$\approx 6,000 \times 3$
- incorporating physical laws as gentle hints, as a softer way of integrating physics compared to PINNs.

Thank you!